



Behavioral Anomaly Detection for Smart Contracts

Hackathon Track: AI Alpha & Data



The Problem

\$3.4B

stolen in crypto hacks in 2025



Signature-based tools
miss novel attacks



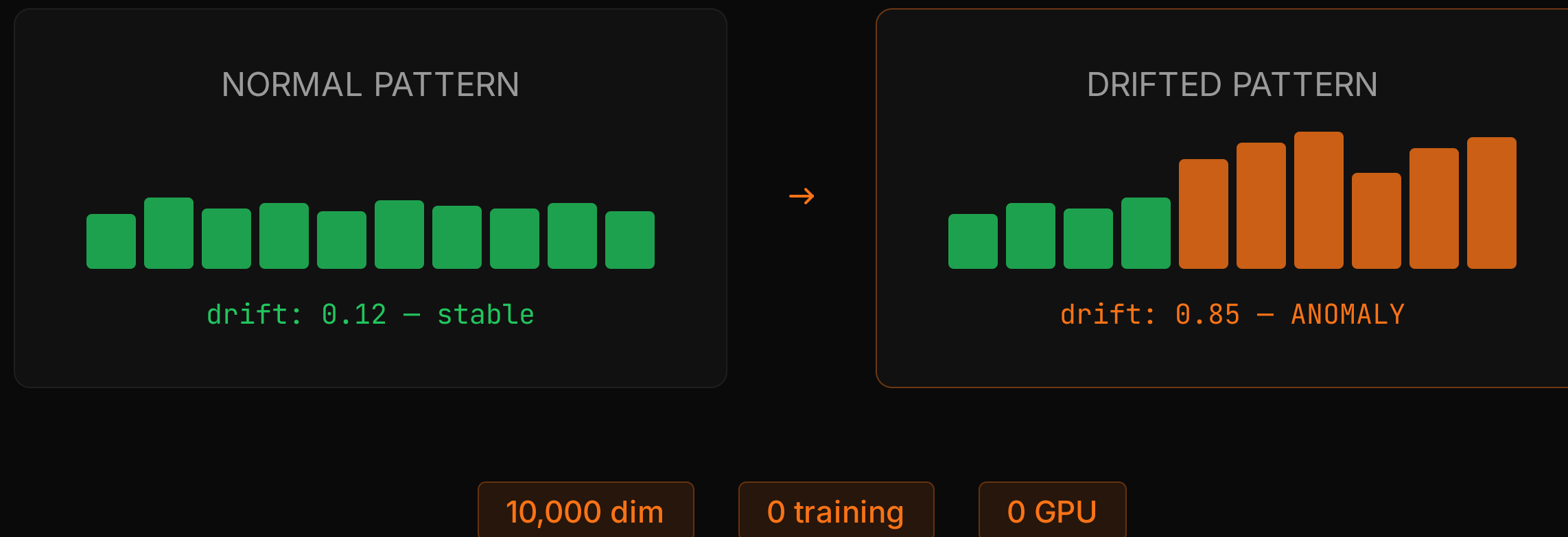
LLM monitors are slow
and non-deterministic



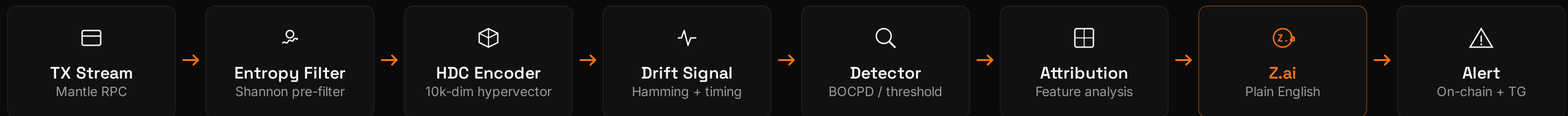
Zero historical data
to train on

Our Approach

Every contract has a behavioral fingerprint.
Sentinel detects the moment it changes.



Pipeline



LLM is **NEVER** in the detection loop — detection is deterministic and reproducible, byte for byte.

Z.ai Integration

STRUCTURED FINDING (INPUT)

feature: gas_p99
drift: 0.847
baseline: 142,000
current: 1,203,000
window: 45-52

Z.ai



PLAIN ENGLISH (OUTPUT)

"Gas consumption spiked 8.5x above baseline in windows 45-52, consistent with a complex multi-call pattern not seen in the warmup period."

Strict template: may only restate Tier-5 findings

No hallucinated severity, no invented causes

Without API key: deterministic fallback (CI-safe)

Live Scan Results



4.3x

separation ratio

0

false positives

129

tests passing

On-Chain Alert Anchoring

SENTINELALERTREGISTRY

0x0899E1507CFfeF8620455721F5bd528Bb072187

Mantle Mainnet · Chain ID 5000



Immutable



Verifiable



Auditable on Mantlescan

Developer Experience



```
$ python -m sentinel scan 0x09bc4e... --min-health 60
```

Health Score: **83 / 100** Healthy

Drift (median): 0.221

Alerts: 4

One command. Full behavioral profile.



GitHub Actions CI gate

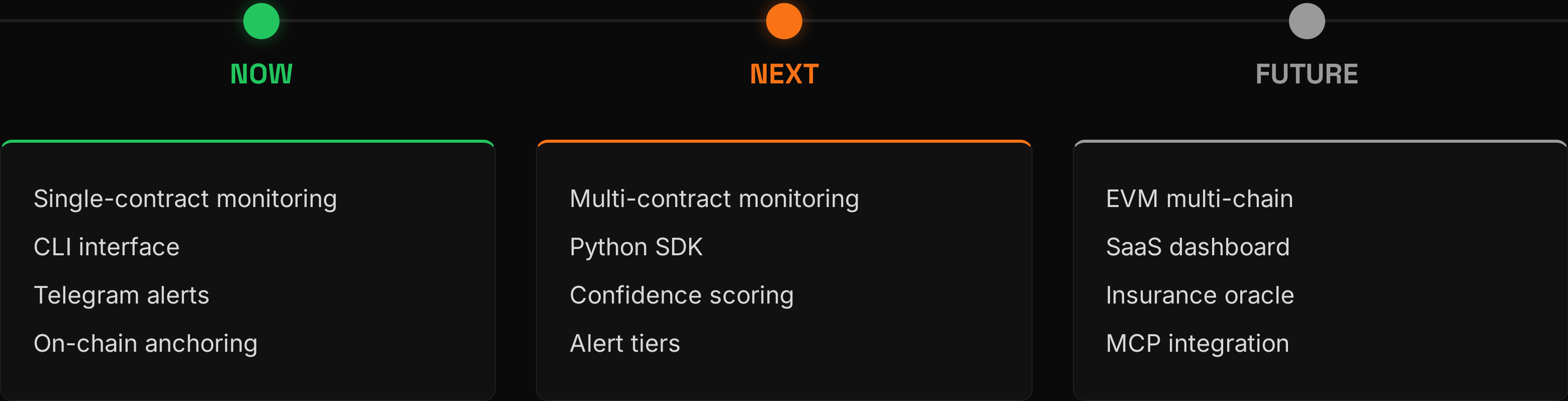


Pre-commit hooks



Telegram alerts

Roadmap





MANTLE SENTINEL

Detect Before They Hit

 mntsentinel.xyz

 mntsentinel.xyz/dashboard

 github.com/alexbelij/mantle-sentinel

 [@MantleSentinelBot](https://twitter.com/MantleSentinelBot)



Scan for GitHub repo